Total No. of Questions : 6]

SEAT No. :

P555

[Total No. of Pages : 4

OCT/BE/Insem.-514A
REFRIGERATION AND AIR CONDITIONING
(2015 Pattern)

Time : 1 Hour] ***[Max. Marks : 30***

Instructions to the candidates :

1) *All questions carry equal marks.*
2) *Use of nonprogrammable scientific calculator is allowed.*
3) *Assume suitable data if necessary.*

Q1) a) How refrigerants are designated? **[4]**

b) Write a short note on Commercial Ice Plant. **[6]**

OR

Q2) a) Write a short note on Cold Storage with respect to refrigerant used and compressor type. **[4]**

b) What are Zeotropes and Azeotropes refrigerants? Write two examples of each. **[6]**

Q3) a) What is the effect of suction superheating on the performance of vapour compression refrigeration system? **[4]**

b) What is an Electrolux refrigeration? **[6]**

OR

Q4) a) Define and explain COP and EER of a VCR system. **[4]**

b) A vapour compression refrigerator uses R40 as refrigerant and operates between temperature limits of –10°C and 45°C. At entry to the compressor, the refrigerant is dry and saturated and after compression it acquires a temperature of 60 “C. Find the COP of the refrigerator.

Take following properties of R-40 refrigerant **[6]**

Temp (°C)	hr (kJ/kg)	hg (kJ/kg)	sr (kJ/kgK)	sg (kJ/kgK)
–10	45.4	460.7	0.183	1.637
45	133	483.6	0.485	1.587

Take Cp of vapour refrigerant at condenser pressure as 1.09 kJ/kgK.

P.T.O.

Q5) a) What are the advantages or multi evaporator system over single evaporator? **[4]**

b) A vapour absorption refrigeration cycle has generator temperature of 120 °C, evaporator temperature of –10 °C and ambient temp of 30°C. Estimate maximum possible COP. The actual COP is 0.5 of the maximum COP. If the capacity of the plant is 100 TR, calculate the fuel consumption of the plant. The calorific value of fuel is 40 MJ/kg. **[4]**

OR

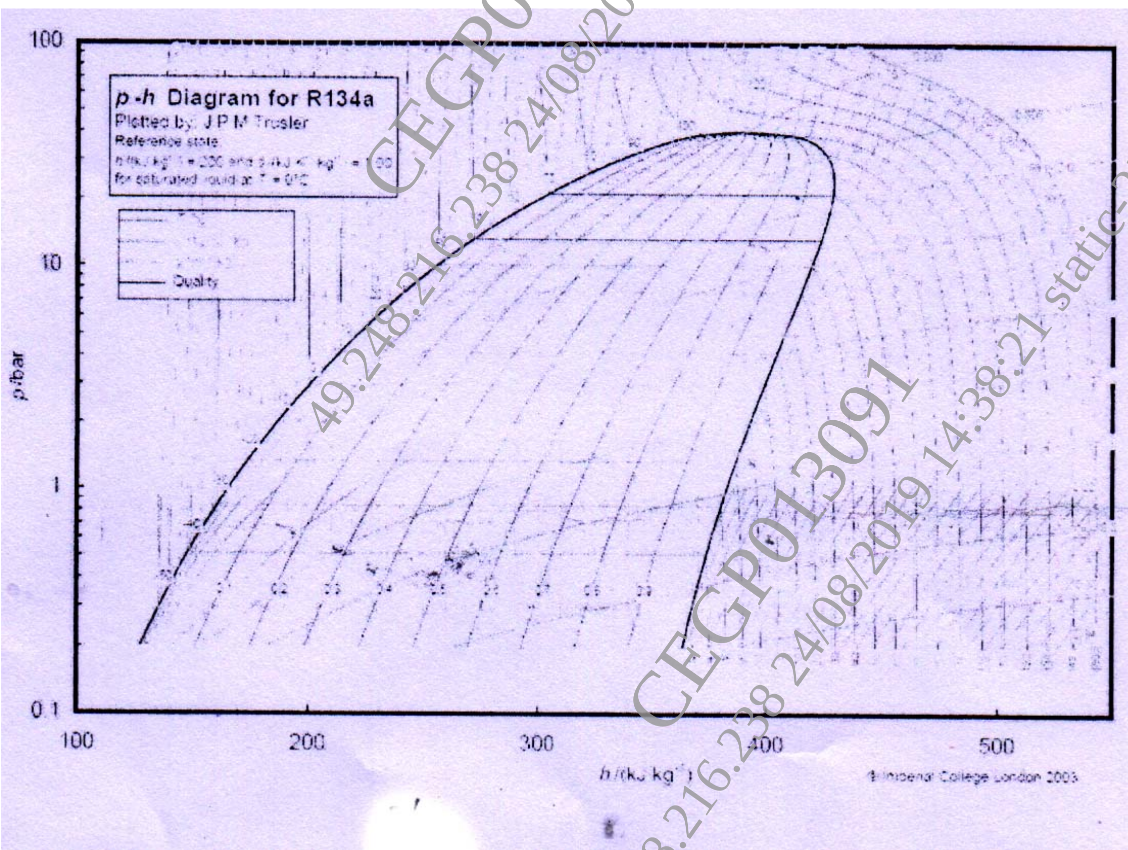
Q6) A single compressor using R134a as refrigerant has three evaporators of capacity 10 TR. 20 TR and 30 TR. All the evaporators operate at –10°C and vapour leaving the evaporator is dry and saturated. The condenser temperature is 40°C. Assuming isentropic compression and no sub-cooling in the condenser. Find **[10]**

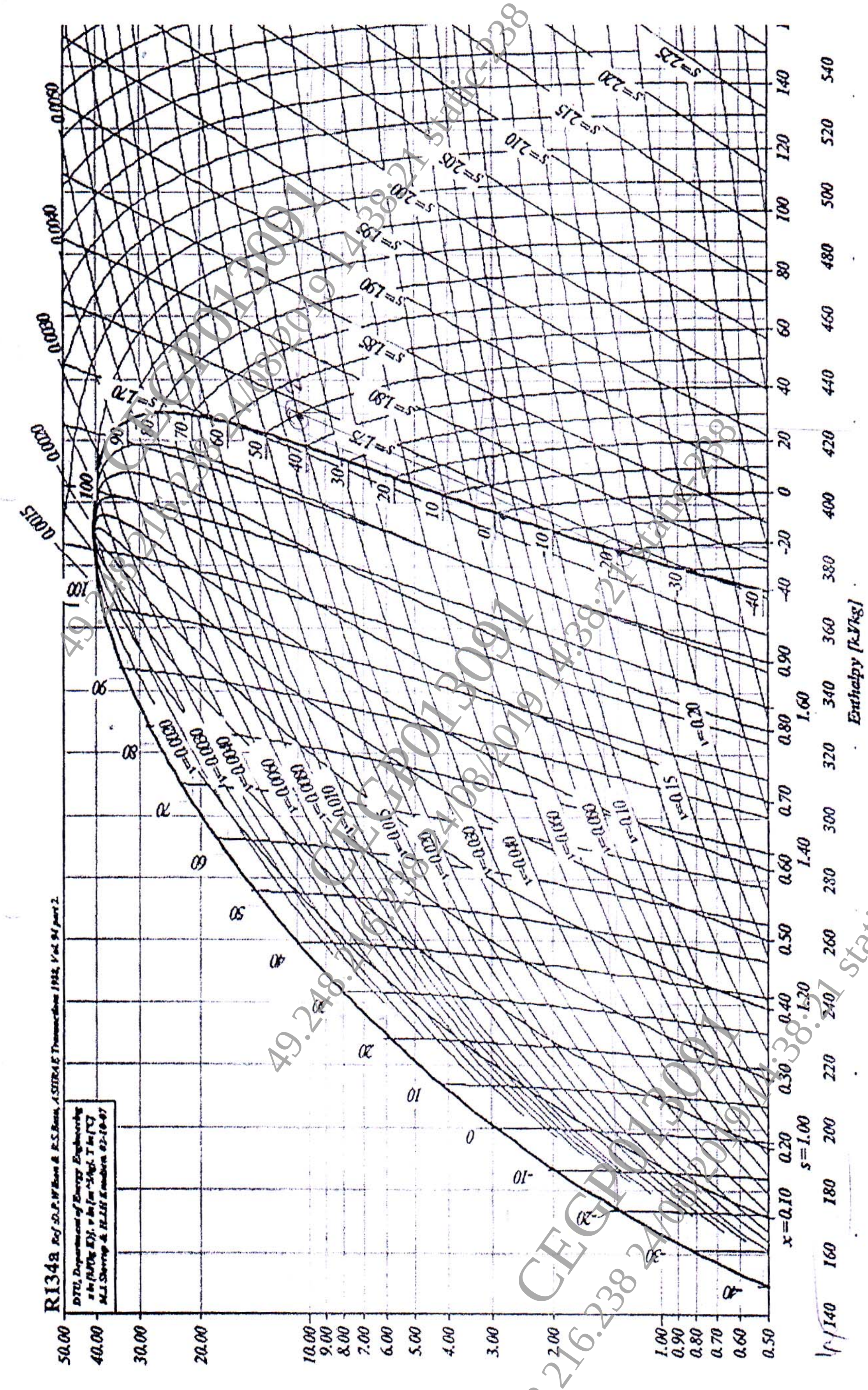
a) Mass flow rate of refrigerant through each evaporator

b) Power required for the compressor

c) COP of the system

Refer p-h chart for R 134a




R134a
Enthalpy [kJ/kg]